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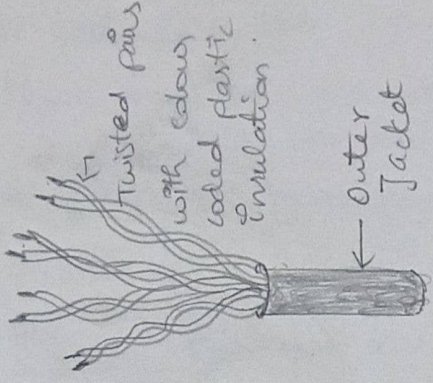
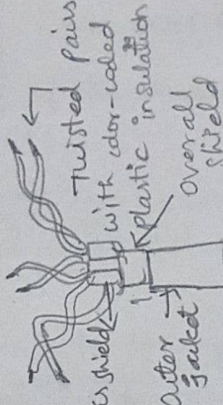
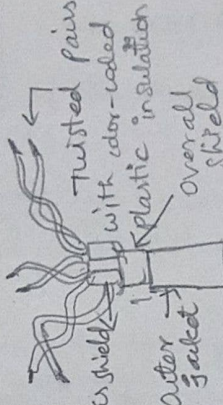
STUDY OF DIFFERENT TYPES OF NETWORK CABLES

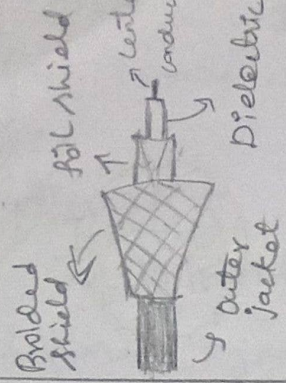
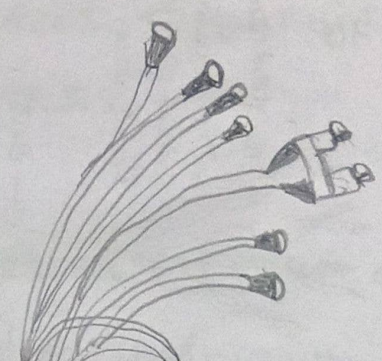
Date: 27/01/20

Aim:-

To understand the different types of network cables.
Different types of cables used in networking cables are:

1. Unshield Twisted Pair (UTP) cable
2. Shield Twisted Pair (STP) cable
3. Coaxial cable
4. Fibre Optic cable.

Cable Type	Category	Maximum Data Transmission	Advantage / Disadvantage	Application / use	Image
UTP	Category 3	10 kbps	Advantages:- * Cheaper in cost * Easy to install as they have a smaller overall diameter. Disadvantages:- * More prone to (EMI) * Electromagnetic interference and noise.	10 Base-T Ethernet, Fast Ethernet, Gigabit Ethernet, Fast Ethernet, Fast Ethernet, Gigabit Ethernet.	
	Category 5	up to 100Mbps			
	Category 5e	1 Gbps			
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * Faster than UTP * Less susceptible to noise and interference.	Gigabit Ethernet, 10 G Ethernet (55 m) widely used in data centres.	
SSTP	Category 7	10 Gbps	Disadvantages:- * Expensive * Greater installation effort.	Gigabit Ethernet, 10 G Ethernet (100m)	

Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages:- * High bandwidth * Faster than UTP * Less susceptible to noise and interference. * Versatile - Disadvantages:- * Limited distance * Cost * Size is bulky.	Speed of signal is 500m Television network High Speed Internet connections.	
Fibre Optic Cable	Single mode multi mode	100 Gbps	Advantages:- * High Speed * High bandwidth * High Security * Long distance. Disadvantages:- * Expensive * Requires Skilled installers	Maximum distance of fibre Optics cable is around 100 metres.	

b) Make your Own Ethernet cross-over cable / Straight cable.

Step 1:- To start construction of the device, begin by threading shields onto the cable.

Step 2:- Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3:- After, you will need to untangle the wires, there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4:- Once the order is correct, bend them together in a line, and if there any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up.

Step 5:- Next, push the cable right in. The notch at the end of the needs to be just over the cable shielding, and if it isn't, that means you stripped off too much shielding.

Step 6:- After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7:- Lastly, repeat for other end using diagram B / using A.

Result:-

The types of network cables and making cross
wired cable to straight through cable using
crimping / crimping tool is studied.